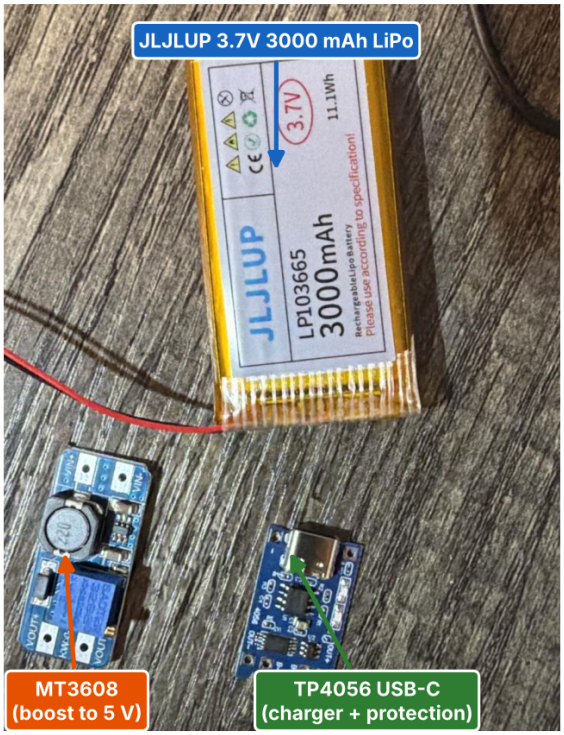
Portable SquachWatch

A battery-powered Cheap Yellow Display build guide



The parts used in this build: 3000 mAh LiPo, MT3608 boost board, TP4056 USB-C charger.

At a glance	
What you're building	A rechargeable battery pack inside a 3D-printed case that runs the SquachWatch firmware on a 2.8" Cheap Yellow Display (ESP32-2432S028R) for about 10 hours, with an on/off switch and USB-C charging.
Two ways to finish it	**Method A — hard-wired:** two wires go straight into the CYD's P1 power connector. Cleanest, no cables in the case. **Method B — USB cable:** a short USB cable (end cut off) runs from the boost board into the CYD's USB port. No connector work on the CYD.
Skill level	Beginner soldering. You'll make about 10 solder joints.
Time	1.5–2 hours, plus ~3.5 hours to charge the battery the first time.
You'll need	Soldering iron, multimeter, wire cutters/strippers, small flat screwdriver, heat source for heat-shrink.
Case files	Free on GitHub: https://github.com/DaGromflomite/squachwatch-cyd-case Downloads (project, STLs, this guide): https://github.com/DaGromflomite/squachwatch-cyd-case/releases/latest

Contents

Section	What's in it
1. Parts list (bill of materials)	Everything to buy or gather, with quantities
2. Tools	What you need on the bench
3. How it works	Why two boards, why no buck converter, runtime math
4. Meet the parts	Annotated photos of each board and the battery
5. Safety + soldering basics	Read before touching the battery
PART 1 — Steps 0–5 (both methods)	Flash SquachWatch → battery → charger → switch → boost board → set 5 V
PART 2 — Method A: hard-wired	Steps A1–A4: P1 power lead, first power-up
PART 3 — Method B: USB cable	Steps B1–B4: cut cable, first power-up
PART 4 — The printed case	Download, pick a plate, print, inserts, badges, assembly
Using it	Charging, the LEDs, flashing new firmware
Troubleshooting + final checklist	Symptom → fix table

HOW TO USE THIS MANUAL: Every step has a [ ] box. Tick it when it's done. If a test in a step fails, stop and use the troubleshooting table before moving on. Every test exists to catch a mistake before it can damage something.

1. Parts list (bill of materials)

Everything needed to build one unit. Links are Amazon searches, because individual listings change constantly; pick a listing whose photos match the description. AliExpress sells all of these too, usually cheaper and slower.

#	Part	Qty	Used for	Where to buy
1	ESP32-2432S028R "Cheap Yellow Display" (CYD), 2.8", resistive touch	1	Runs SquachWatch	Amazon search
2	3.7 V 3000 mAh LiPo **with built-in protection**, size 103665 (10 × 36 × 65 mm). Built with a JLJLUP LP103665; any same-size protected cell works.	1	Power	Amazon search
3	TP4056 USB-C charger module **with protection** (DW01 + 8205A chips, pads OUT+ B+ B– OUT–)	1	Charges the battery; protects it	Amazon search
4	MT3608 DC-DC boost module (VIN+ VIN– VOUT+ VOUT–, blue trimmer)	1	Raises 3.7 V to 5 V	Amazon search
5	SS12D00 mini slide switch (SPDT, 3 pins)	1	On/off	Amazon search
6	Silicone hook-up wire 24–26 AWG, red + black	~50 cm each	Board-to-board wiring	Amazon search
7	Heat-shrink tubing 2 mm and 3 mm	~10 pcs	Insulating joints	Amazon search
8	Kapton (polyimide) tape	1 roll	Insulating boards	Amazon search
9	Double-sided foam tape	small	Mounting parts	Amazon search
10	**Method A only:** 4-pin cable that fits the CYD's P1 socket (1.25 mm pitch, often sold as MX1.25 / "JST 1.25" 4-pin). Most CYDs ship with one in the box; if yours did, use it.	1	P1 power lead	Amazon search
11	**Method B only:** short USB cable with the plug the CYD takes (micro-USB or USB-C, depending on board version), one you don't mind cutting	1	Power into CYD USB	Amazon search
12	3D-printed case: free files on GitHub (see PART 4)	1	Enclosure	GitHub

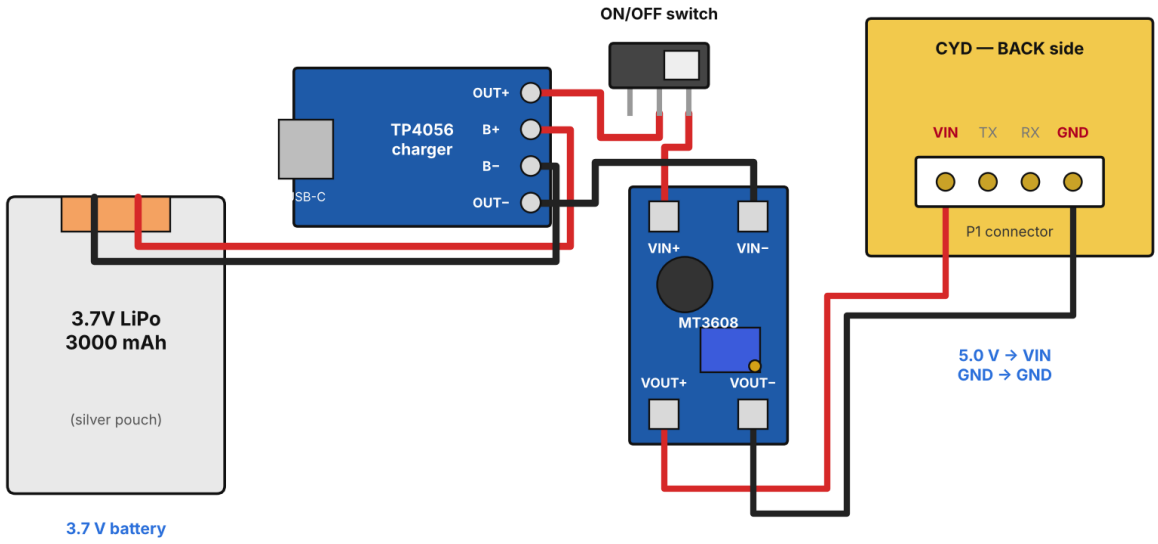
#	Part	Qty	Used for	Where to buy
13	**M4 × 6 × 6 mm brass heat-set threaded inserts** (M4 thread, 6 mm long, 6 mm outside diameter, knurled). They melt into the 5.7 mm holes in the case corners.	4	The case screws thread into these	Amazon search
14	**M4 × 8 mm button-head socket screws** (ISO 7380, hex drive), stainless or black. M4 × 6 mm also works but grips less thread.	4	Hold the bezel on	Amazon search
15	CA (super) glue	a few drops	Gluing in the two badges	Amazon search

COST: About $25–40 total, including the CYD (roughly $12–20). The battery, charger, boost board and switch are about $10–15 together.

2. Tools

Tool	Notes
Soldering iron + solder	Any adjustable iron. 330 °C with leaded solder, 350 °C with lead-free. Fine conical or small chisel tip.
Flux (pen or paste)	Makes every joint easier. Optional but strongly recommended.
Multimeter	Needed for: battery check, setting 5 V, polarity, continuity.
Flush cutters + wire strippers	For 24–26 AWG.
Small flat screwdriver (2 mm)	To turn the MT3608's brass screw.
Heat gun or lighter	For heat-shrink. Keep flame away from the battery.
Helping hands / tape	To hold boards while soldering.
Tweezers	Optional, handy.

3. How it works



The complete circuit (Method A shown). Power flows left to right.

The power chain, one part at a time

Part	What it does	Voltage in → out
Battery	Stores energy.	— → 3.0 to 4.2 V (3.7 V nominal)
TP4056	**Charges** the battery from USB-C, and **protects** it: cuts power if it drains too low (~2.5 V), is overcharged, or is shorted.	Battery voltage passes straight through to OUT+/OUT–
Switch	Connects/disconnects everything after the charger.	same
MT3608	**Boosts** the battery voltage up to a steady 5 V.	3.0–4.2 V → 5.0 V
CYD	Runs on 5 V from its P1 VIN pin (Method A) or its USB port (Method B).	5 V

Do I need a buck converter?

No. A *buck* converter lowers voltage. The battery is *below* the 5 V the CYD needs, so you need to *raise* it, which is a *boost* converter. The MT3608 is your boost converter.



A BUCK converter steps voltage DOWN, so it is the wrong part here. The MP2636 board IS the boost converter.

DON'T: Never connect the battery (3.7 V) directly to the CYD. It's too low for the 5 V input; the CYD would reboot or not start.

Why the switch sits *after* the charger

The switch is in the wire from TP4056 **OUT+** to MT3608 **VIN+**. With the switch OFF, the CYD and boost board are disconnected but the battery is still connected to the charger, so **you can charge with the device switched off**, which is the best way to charge (see "Using it").

How long will it run?

Worst case for SquachWatch (WiFi/BLE scanning, screen on): 200 mA at 5 V. The MT3608 is about 88% efficient at this load.

```
Runtime = (battery Wh × efficiency) ÷ (5 V × current in A)
Runtime = (11.1 × 0.88) ÷ (5 × 0.2)

Multiply the top (11.1 × 0.88):
Runtime = 9.77 ÷ (5 × 0.2)

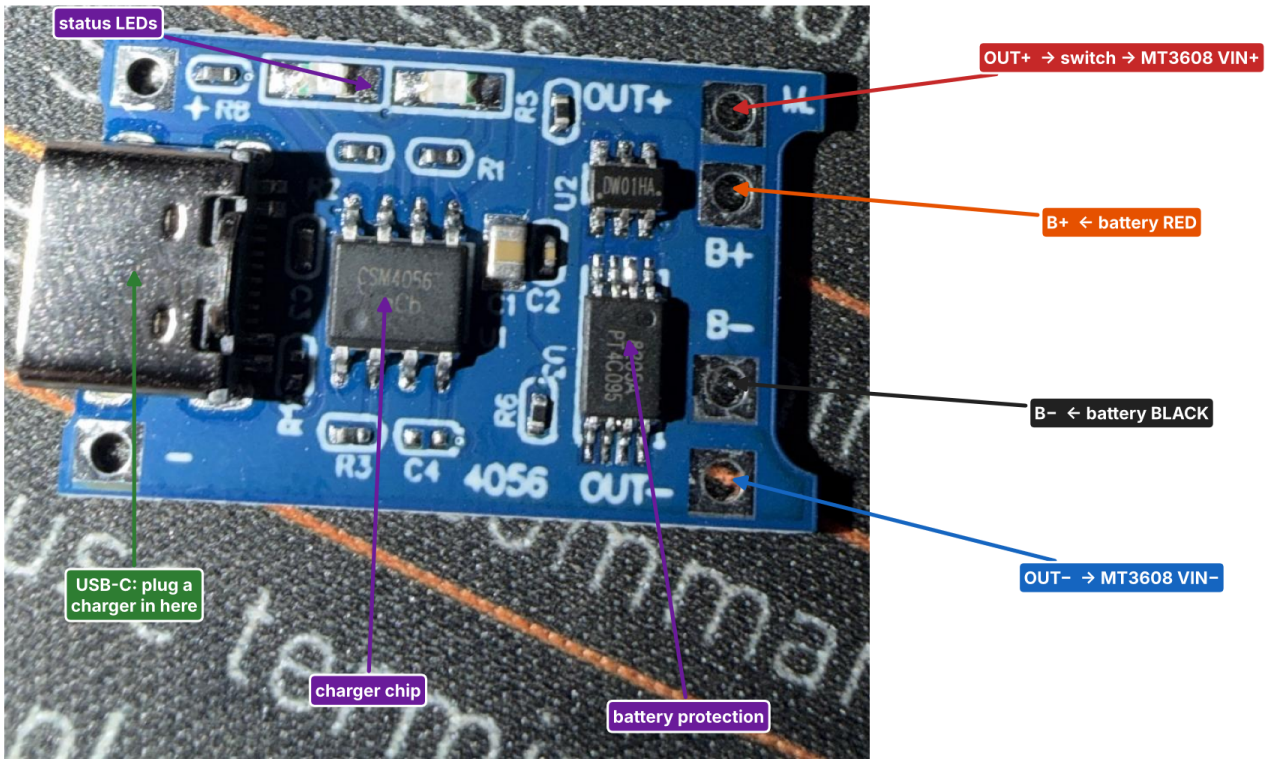
Multiply the bottom (5 × 0.2):
Runtime = 9.77 ÷ 1.0

Divide:
Runtime ~ 9.8 hours
```

TIP: Anything else plugged into the CYD (LED strips or matrices on CN1, sensors on P3) draws power too. A 64-LED WS2812 matrix alone draws 50–60 mA even when dark, costing about 2 hours of runtime. Unplug add-ons you don't need.

4. Meet the parts

The TP4056 USB-C charger (with protection)

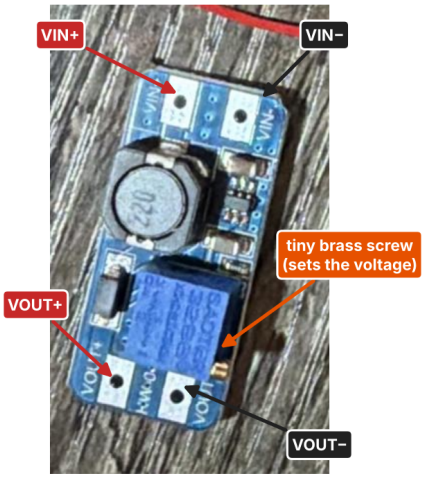


The USB-C TP4056 used in this build. Pads run down the right edge in this order: OUT+, B+, B–, OUT–. The label sits next to or just below each hole, so read carefully. Other TP4056 variants may order the pads differently; always go by the printed labels.

Pad	Connects to	Notes
B+	Battery **red**	Battery positive.
B–	Battery **black**	Battery negative.
OUT+	Switch → MT3608 VIN+	The load's positive. (Electrically the same as B+.)
OUT–	MT3608 VIN–	The load's negative. Goes through the protection chip, which is why the load must use OUT–, **never B–**.
+ / – (left side)	Nothing	Alternate charge input. Not used; USB-C does the same job.

LEDs: one lights (usually red) while charging; the other (usually blue or green) lights when full. **Charge rate:** about 1 A, so a 3000 mAh battery takes ~3.5 hours from empty.

The MT3608 boost board



Pad	Connects to
VIN+	Switch (from TP4056 OUT+)
VIN–	TP4056 OUT–
VOUT+	CYD 5 V (P1 VIN, or the USB cable's red wire)
VOUT–	CYD GND (P1 GND, or the USB cable's black wire)
Brass screw	Sets the output voltage. **Must be set to 5.0 V before anything is connected to VOUT (Step 5).**

DANGER: MT3608 boards ship set to a random voltage, sometimes 12–28 V. Wired to the CYD before you adjust it, it will destroy the CYD instantly. Step 5 is not optional.

The battery



Spec	Value
Model used	JLJLUP LP103665 (any protected 103665 cell works)
Size	10 × 36 × 65 mm (that's what 103665 means)
Voltage	3.7 V nominal; 4.2 V full; ~3.0 V empty
Capacity	3000 mAh / 11.1 Wh
Protection	Built-in PCB under the tape at the wire end, plus the TP4056's own protection. Two layers.

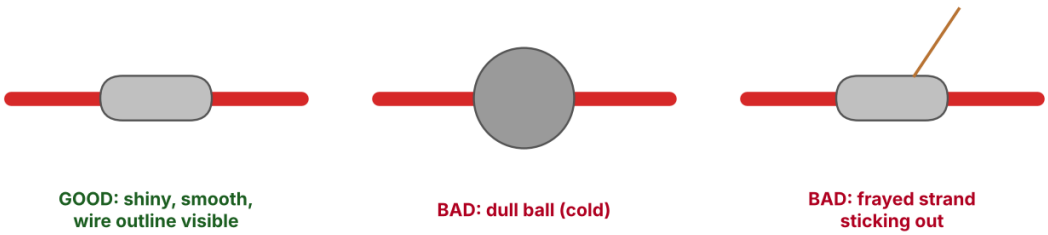
5. Safety + soldering basics

LiPo safety rules

1	Work on a non-flammable surface (tile, metal tray, silicone mat). Keep the iron's tip away from the battery pouch at all times.	[ ]
2	Never let the battery's red and black wires touch each other. If you have to cut or strip them, do **one wire at a time** and tape the bare end before touching the other.	[ ]
3	Never puncture, bend, crush or cut into the battery or the tape over its protection PCB.	[ ]
4	A battery that is puffy, swollen, hot, or smells sweet is done. Put it outside on concrete and take it to battery recycling.	[ ]
5	Only charge it through the TP4056. Never plug a phone charger into the battery wires directly.	[ ]
6	Don't leave it charging unattended the first few times. Feel it now and then: warm is OK, hot is not.	[ ]

Soldering in 5 moves

1	**Tin the pad:** touch the iron to the pad for 1–2 s, feed a little solder in. It should flow and look shiny.	[ ]
2	**Tin the wire:** strip 4–5 mm, twist the strands tight, heat the wire and feed solder in until it soaks through.	[ ]
3	**Join:** push the tinned wire through the pad hole (or lay it on the pad), touch the iron to both for 2 s until the solder melts together. Remove the iron, hold still for 3 s.	[ ]
4	**Inspect:** shiny, smooth, wire outline visible = good. Dull ball or grainy = cold joint; add flux, reheat.	[ ]
5	**Insulate:** heat-shrink over wire joints; Kapton tape over board pads that could touch something.	[ ]



Left: a good joint. Middle: a cold joint (reheat with flux). Right: a stray strand (trim it, it can short).

PART 1 — Both methods

Steps 0–5 are identical for Method A and Method B. Don't skip ahead.

Step 0: Flash SquachWatch onto the CYD (before any wiring)

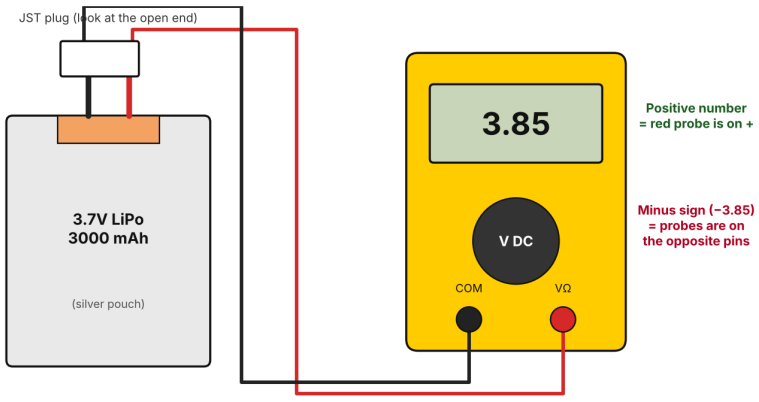
Do this first, with the CYD on its own and powered only by the USB cable. It proves the board and screen work before you solder anything, and it's the only time the CYD is easy to reach.

0.1	On a desktop or laptop, open **squachwatch.com** in Chrome, Edge, Brave or Firefox (the SquachWatch README lists these). Phones and tablets won't work.	[ ]
0.2	Plug the CYD into the computer with a USB **data** cable. Some cables only charge; if nothing shows up, try another cable.	[ ]
0.3	In the board list, pick **2.8" CYD · ST7789**. Not sure which board you have? The flasher has a box for the code printed on the back of the board.	[ ]
0.4	Click **Connect & Install**, choose the CYD's serial port in the pop-up, and wait for it to finish. Don't unplug it while it's writing.	[ ]
0.5	Optional: click **Set Time & Zone** right after flashing so the clock is correct.	[ ]
0.6	The CYD restarts into SquachWatch. **Solid white screen?** Your board has the other screen type: flash again and pick **2.8" CYD · ILI9341**.	[ ]
0.7	**Port doesn't appear at all?** Many CYDs use a CH340 USB chip that needs a driver on Windows. Install the CH340 driver, then replug.	[ ]
0.8	Unplug the USB cable. You'll power it from the battery from now on. To update later, see "Using it" → Flashing new firmware.	[ ]

LATER: Never have the battery switched ON while the CYD is also plugged into a computer. Switch OFF first, every time you plug in USB.

Step 1: Check the battery

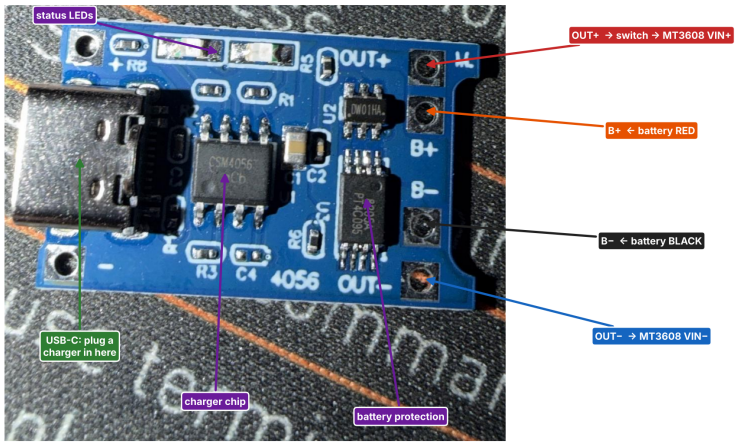
Two questions: is the battery healthy, and which wire is really positive?



Measuring battery voltage. If your battery has bare wire ends instead of a plug, touch the probes to the bare wires the same way.

1.1	Meter dial to **DC volts** (V with a straight line, not the wavy one), **20 V** range. Black lead in **COM**, red lead in **VΩ**.	[ ]
1.2	Touch the **red probe to the battery's red wire** (or its contact in the plug) and the **black probe to the black wire**.	[ ]
1.3	Read the screen. **3.6–4.2 V, no minus sign** = healthy and the colors are correct. Write the number here: _____ V	[ ]
1.4	**Minus sign?** The colors are swapped on this battery. Put a piece of tape on the wire that touched the black probe and write "+" on it. From now on, that's your positive wire whatever its color.	[ ]
1.5	**Below 3.0 V?** Don't use it. **0 V?** The battery's protection may have tripped; it may recover once connected to the TP4056 and charged (Step 3). If it doesn't, return it.	[ ]
1.6	**Does it have a plug on the end?** Cut it off as close to the plug as possible, **one wire at a time**. Strip 4 mm off each wire and tin it. Put a small piece of tape over each bare end until you solder it.	[ ]

Step 2: Battery → TP4056



The B+ and B– holes are the middle two on the right edge.

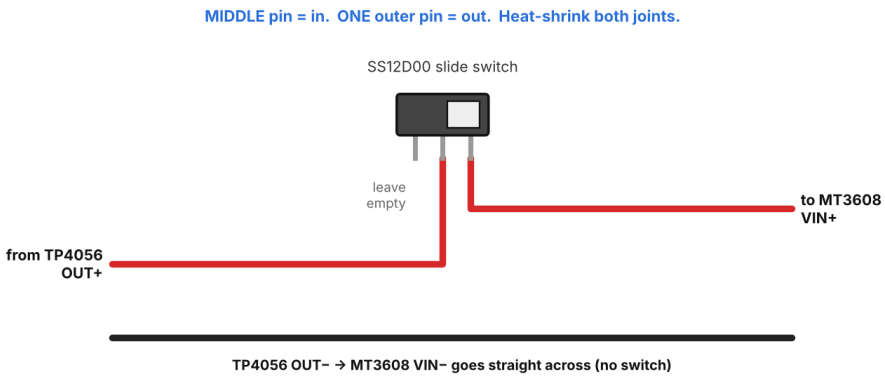
2.1	Lay the TP4056 component-side up, USB-C on the left, so it looks like the photo.	[ ]
2.2	Tin the **B+** and **B**– holes.	[ ]
2.3	Slide 1 cm of heat-shrink onto the red wire, pushed well back.	[ ]
2.4	Solder the battery's **red (+)** wire into **B+**. Feed it through from the top and solder on the bottom, or lay it on the pad.	[ ]
2.5	Tape the joint, then solder the battery's **black (–)** wire into **B**–.	[ ]
2.6	Inspect: no solder bridge between B+ and B– (they're close). No stray strands.	[ ]
2.7	**Test:** meter on DC volts, red probe on **OUT+**, black on **OUT**–. You should read the same battery voltage as Step 1. If you read 0 V, plug a USB-C charger in for a few seconds; that resets the protection. Check again.	[ ]

PASS: Reading the battery voltage on OUT+/OUT– means the battery, both joints and the protection circuit are all working.

Step 3: First charge

3.1	Plug a USB-C phone charger (5 V, 1 A or more) into the TP4056.	[ ]
3.2	The **charging LED** lights (usually red). When it switches to the other LED (blue/green), it's full.	[ ]
3.3	Feel the battery and board after 10 minutes: warm is fine, **hot** means unplug immediately and recheck Step 2.	[ ]
3.4	Let it reach full (up to ~3.5 h) before continuing, or at least while you work on the next steps. Full = 4.15–4.2 V on OUT+/OUT–.	[ ]

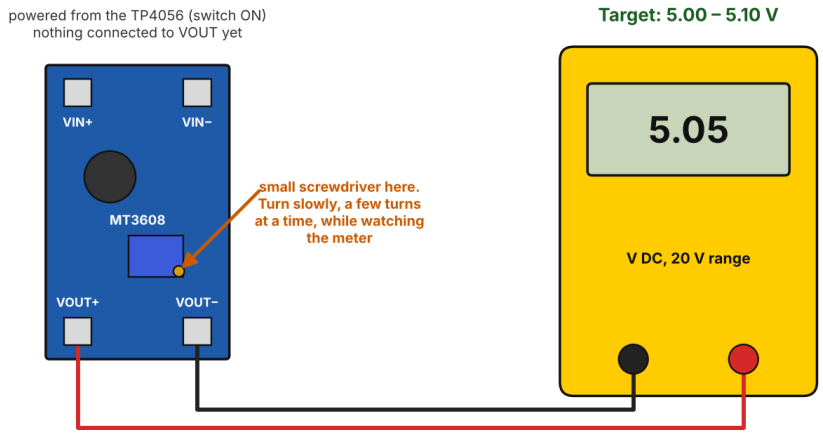
Step 4: Switch + TP4056 → MT3608



The switch interrupts only the positive wire. The negative wire runs straight from OUT– to VIN–.

4.1	Unplug the USB-C charger.	[ ]
4.2	Cut three wires: a red ~6 cm (OUT+ → switch), a red ~6 cm (switch → VIN+), a black ~8 cm (OUT– → VIN–). Adjust lengths to your case layout (PART 4). Strip 4 mm off every end and tin them.	[ ]
4.3	Slide heat-shrink onto both red wires now.	[ ]
4.4	Solder red wire #1 from TP4056 **OUT+** to the switch's **middle pin**.	[ ]
4.5	Solder red wire #2 from **one outer pin** of the switch to MT3608 **VIN+**. Leave the third pin empty.	[ ]
4.6	Shrink the tubing over both switch joints so no metal is showing.	[ ]
4.7	Solder the black wire from TP4056 **OUT**– to MT3608 **VIN**–.	[ ]
4.8	**Leave MT3608 VOUT+ and VOUT– completely empty for now.**	[ ]
4.9	**Test:** slide the switch one way. Meter red on VIN+, black on VIN–: you should see battery voltage (3.7–4.2 V). Slide it the other way: 0 V. Mark the ON direction with a marker.	[ ]

Step 5: Set the MT3608 to 5.0 V (the most important step)



Meter on VOUT+ / VOUT–, nothing else connected, while you turn the brass screw.

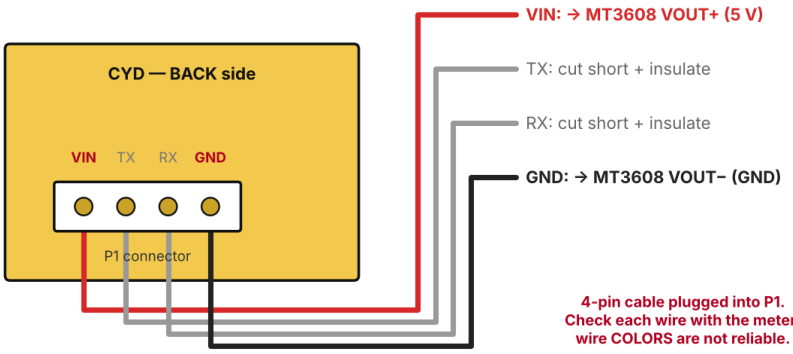
5.1	Make sure **nothing** is connected to VOUT+ or VOUT–.	[ ]
5.2	Switch **ON**.	[ ]
5.3	Meter on **DC volts, 20 V range** (if your meter auto-ranges, any DC volts setting). Red probe on **VOUT+**, black on **VOUT**–. Note the reading: _____ V. It can be anything from 4 V to over 20 V.	[ ]
5.4	Put the small screwdriver in the **tiny brass screw** on the side of the blue box. Turn it **counter-clockwise** about 5 full turns, then check the meter.	[ ]
5.5	If the voltage **went down**, keep turning counter-clockwise, a few turns at a time. If it **went up**, you're going the wrong way on this board; turn clockwise instead.	[ ]
5.6	**Don't panic if nothing changes for 10–20 turns.** These trimmers have about 25 turns end to end and many arrive turned all the way to one end. Keep going.	[ ]
5.7	As you get close, slow down to quarter turns. Stop at **5.00–5.10 V**.	[ ]
5.8	Switch OFF, switch ON, and measure again. It must come back at 5.00–5.10 V.	[ ]
5.9	Put a dab of nail polish or hot glue on the brass screw so it can't get bumped in your pocket.	[ ]

WHY SO PRECISE: Below ~4.8 V the CYD may reboot when WiFi starts. Above 5.5 V you risk damaging it. 5.05 V is the sweet spot.

STUCK? Can't get it below 5 V? On some boards the minimum is about 5 V with no load, or the trimmer is faulty. If it won't go lower than 5.3 V, replace the MT3608; they're about $1 each. **Won't change at all?** Check that VIN has battery voltage (Step 4 test).

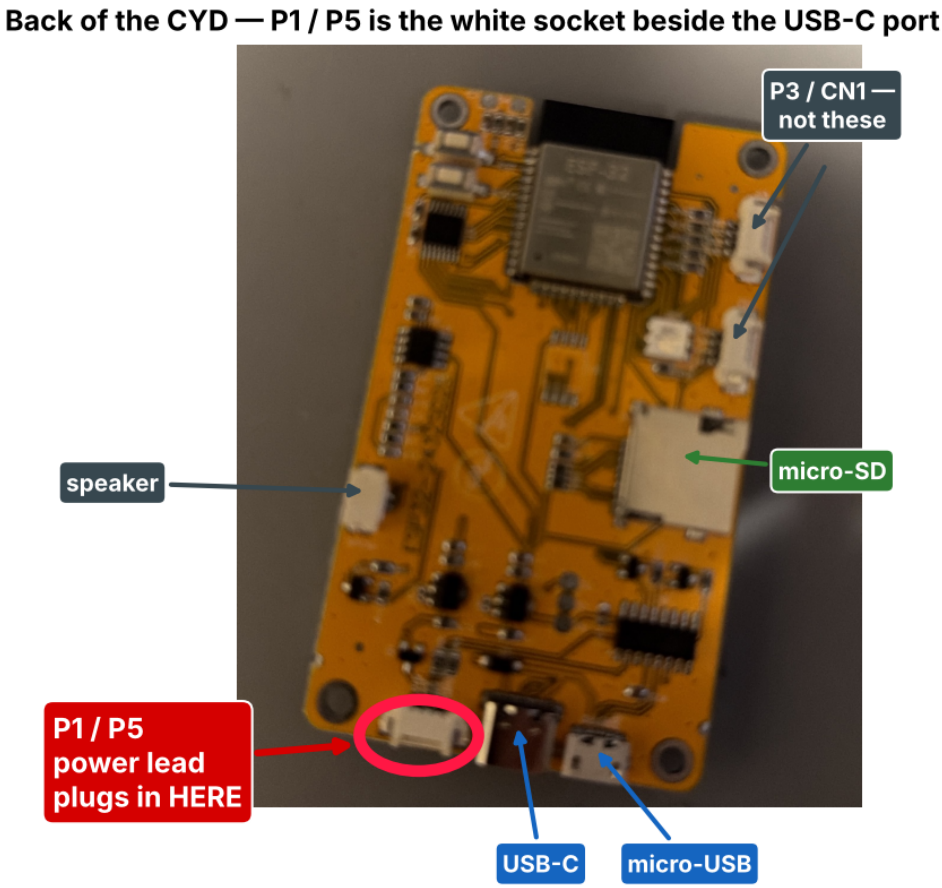
PART 2 — Method A: hard-wired to P1

Two wires go from the MT3608's output into the CYD's **P1** connector (printed **P5** on newer dual-USB boards), the 4-pin socket on the back of the CYD labeled **VIN · TX · RX · GND**. VIN is the CYD's 5 V power input (the same rail USB feeds). You only use VIN and GND. **Skip to PART 3 if you're doing Method B.**



A 4-pin P1 cable becomes the power lead. Verify every wire with the meter.

Step A1: Make the P1 lead



Where P1 is: the white 4-pin socket on the USB end, right next to the USB-C port (labeled P5 on dual-USB boards). The two similar sockets on the long edge (P3 and CN1) are NOT the power input.

P1 or P5? Older single-USB CYDs print **P1** next to this socket. Newer boards with both USB-C and micro-USB print **P5** instead. It's the same 4-pin power/serial socket either way. This guide calls it P1. **Before wiring, confirm which pin is VIN and which is GND** by reading the tiny labels next to the pins, or with the meter test below.

Identify VIN and GND with the meter (if the labels are unreadable)

ID1	Power the CYD from USB. Nothing plugged into P1/P5.	[ ]
ID2	Meter on **DC volts, 20 V range**. Hold the **black probe on the metal shell of the CYD's USB-C port** (that's ground).	[ ]
ID3	Touch the **red probe** to each of the socket's 4 solder joints on the back of the board, one at a time.	[ ]
ID4	**About 4.7–5.0 V = VIN. 0 V = GND.** The other two (TX, RX) read around 3.3 V or jump around.	[ ]
ID5	Write down which end is VIN and which is GND, then unplug the USB.	[ ]
A1.1	Find **P1** (printed **P5** on dual-USB boards): turn the CYD over. It's the **white 4-pin socket on the same short edge as the USB ports, right beside the USB-C port** (photo above). Use the labels or the meter test above to know which end is VIN and which is GND. **Not CN1** or **P3**, the two GPIO sockets that look the same.	[ ]
A1.2	Plug the **4-pin cable** into **P1**.	[ ]
A1.3	Meter on **continuity** (the beep / sound-wave symbol). Touch one probe to the **VIN** pin's solder joint on the back of the CYD, and the other probe to each DuPont end in turn. The one that beeps is VIN. Wrap red tape on it.	[ ]
A1.4	Do the same for **GND**. Wrap black tape on it.	[ ]
A1.5	Unplug the cable from the CYD.	[ ]
A1.6	If the other end has a connector or DuPont pins, cut them off all four wires.	[ ]
A1.7	Cut the **TX** and **RX** wires back to about 2 cm and cover each end with heat-shrink, so they can never touch anything.	[ ]
A1.8	Strip 4 mm from the VIN and GND wires, twist, tin. Slide heat-shrink onto each.	[ ]

Step A2: Solder the lead to the MT3608

A2.1	Switch **OFF**.	[ ]
A2.2	Solder the **VIN** wire (red tape) to MT3608 **VOUT+**.	[ ]
A2.3	Solder the **GND** wire (black tape) to MT3608 **VOUT**–.	[ ]
A2.4	Inspect for bridges and stray strands.	[ ]

Step A3: Check before you plug in

A3.1	CYD still unplugged. Switch **ON**.	[ ]
A3.2	Meter on DC volts. Put the red probe into the JST plug's **VIN** contact and the black into its **GND** contact (a sewing pin or stripped wire helps reach inside the plug).	[ ]
A3.3	Must read **5.00–5.10 V, no minus sign**. Minus sign = VIN and GND are swapped in your lead; fix it before going on.	[ ]
A3.4	Switch **OFF**.	[ ]

Step A4: First power-up

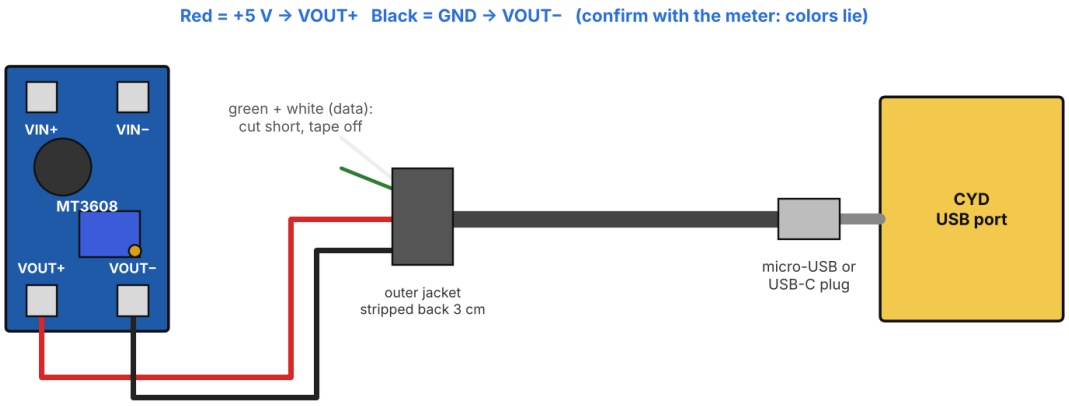
A4.1	Make sure the CYD's USB port is **not** connected to anything.	[ ]
A4.2	Plug the lead into the CYD's **P1**.	[ ]
A4.3	Switch **ON**. SquachWatch boots: splash screen, then the matrix rain.	[ ]
A4.4	Leave it running 15 minutes. The MT3608 gets warm; the battery should stay barely warm. Anything **hot** → switch off, check the troubleshooting table.	[ ]
A4.5	Re-measure VOUT with it running: it should still be 4.9–5.1 V.	[ ]

RULE: Method A: **always switch the battery OFF before plugging the CYD into a computer** to flash it. With both connected, the MT3608 and the USB port both drive the same 5 V rail.

NEXT: Done with Method A. Go to **PART 4**.

PART 3 — Method B: USB cable

The MT3608 has no USB socket, so you make one: cut a USB cable, keep the end that plugs into your CYD, and solder its power wires to VOUT. Some CYD versions have two USB ports (micro-USB and USB-C); either works.



Only the power pair (normally red + black) is used. The data pair is cut off.

Step B1: Prepare the cable

B1.1	Pick a cable with the plug your CYD takes. Cut it so you keep the CYD-plug end plus about 10–15 cm of cable (enough to reach in your case).	[ ]
B1.2	Carefully slit and remove 3 cm of the outer jacket. Don't nick the wires inside. You'll find 4 wires (sometimes a foil shield and bare drain wire too).	[ ]
B1.3	Usually: **red = +5 V**, **black = GND**, green and white = data. **Cheap cables ignore these colors**, so verify in the next step.	[ ]
B1.4	Cut the shield/drain wire and the two data wires flush with the jacket.	[ ]

Step B2: Identify the wires with the meter

B2.1	Meter on **continuity**.	[ ]
B2.2	Look into the plug: the **VBUS (+5 V)** and **GND** contacts are the outermost ones in micro-USB; in USB-C they're the wide pins at both ends of each row. Easiest method: touch one probe to the **outside metal shell** of the plug, then try each wire. The one that beeps is **GND** (the shell is grounded). Mark it with black tape.	[ ]
B2.3	The remaining power-colored wire (normally red) is +5 V. Mark it with red tape.	[ ]
B2.4	Strip 4 mm, twist, tin. Slide heat-shrink on.	[ ]

Step B3: Solder to the MT3608

B3.1	Switch **OFF**.	[ ]
B3.2	**+5 V wire** (red tape) → MT3608 **VOUT+**.	[ ]
B3.3	**GND wire** (black tape) → MT3608 **VOUT**–.	[ ]
B3.4	Inspect, then switch ON and measure: red probe on VOUT+, black on VOUT–: still 5.00–5.10 V. Switch OFF.	[ ]

Step B4: First power-up

B4.1	Plug the cable into the CYD.	[ ]
B4.2	Switch **ON**. SquachWatch boots.	[ ]
B4.3	Run 15 minutes; check nothing is hot.	[ ]
B4.4	**To flash new firmware later:** switch OFF, unplug this internal cable from the CYD, plug your computer in instead. No back-feed risk with this method.	[ ]

TIP: Use a cable with **right-angle** plug if you can find one; a straight plug sticks out ~2 cm and can force your case to be bigger.

PART 4 — The printed case

The case is free to download from GitHub: https://github.com/DaGromflomite/squachwatch-cyd-case. It's a two-part case (back shell + screen bezel) for **Method A** (hard-wired to P1/P5). Method B's internal USB cable doesn't fit. Outside size: **105 × 62.7 × 24.4 mm** including the stylus tube.

What you want	Where to get it
Bambu Studio project, 3 plates, colors set (easiest)	squachwatch_case_v20.3mf
STL files for any slicer	STL folder or the STL zip on the Releases page
This guide (latest version)	Build-Guide folder
Change the design	source folder (Python; every dimension is a parameter)

USB-C charge port + switch side

SD-card side + CYD USB end



Back shell (blue) and screen bezel (orange). No supports needed.

Pick a plate

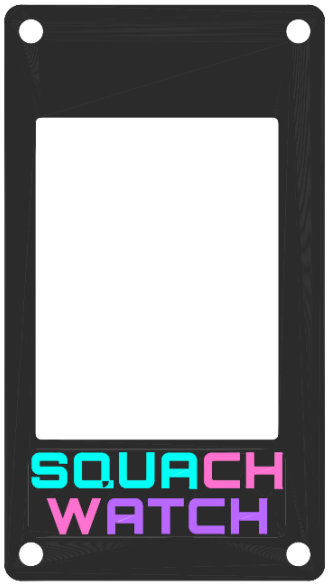
Plate	What's on it	Pick it if you hold it…
1: Horizontal	Shell, plain bezel, Squachy badge, SQUACHWATCH badge. The lettering goes on the **back**, above Squachy	Sideways (landscape), USB end on the right. This is SquachWatch's default screen rotation
2: Plain	Shell and bezel, no pockets, no badges	Any way; one color
3: Vertical	Shell, bezel, Squachy badge, SQUACHWATCH badge. The lettering sits on the bezel under the screen, in two big lines (SQUACH / WATCH)	Upright (portrait), USB cable at the bottom

v18_stylus: SQUACHWATCH nameplate badges

HORIZONTAL: back of the case (turned over)



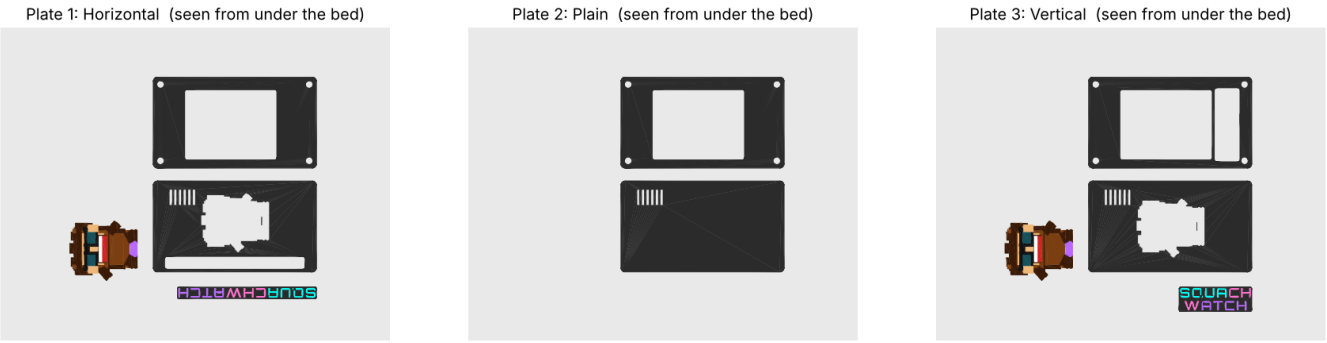
VERTICAL: front of the case



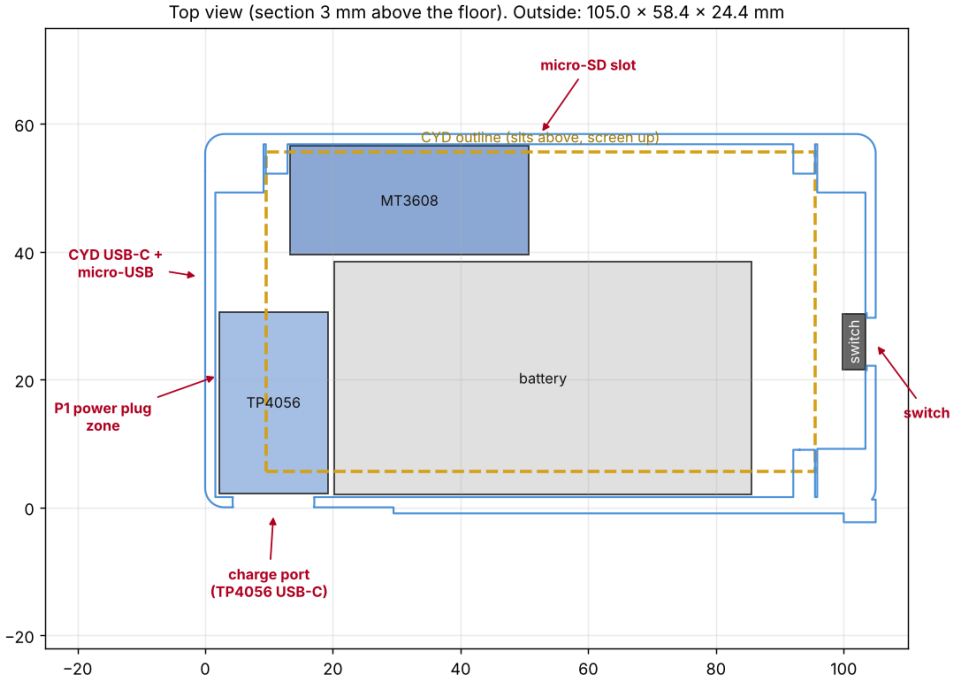
Left: the back of the Horizontal case. Right: the front of the Vertical case.

Printing

Item	Detail
Material	PETG or ASA (it sits against a warm battery); PLA is fine indoors
Settings	0.2 mm layers, 4 walls, 20 % gyroid, **no supports**
Brim	Not needed (the project file has it off)
Orientation	Already set in the files: shell open-side up; bezel and both badges face down
Colors	Plates 1 and 3 use 11 filaments (case color, 8 Squachy colors, cyan/pink/purple lettering). Map them to your AMS when you print. Plate 2 is one color
Openings	CYD USB-C + micro-USB (USB end), TP4056 USB-C charge port (long wall, under the stylus tube: a 15 × 9.6 mm notch that fits the cable's plug body), switch slot (far end), micro-SD slot with fingernail scoop (long wall), speaker grille (floor), stylus tube (outside the charge-port wall)
Badges	The Squachy badge and the SQUACHWATCH badge print as separate pieces, 1.1 mm thick. Each fits a 0.6 mm pocket and stands 0.5 mm proud, so you can feel it. The SQUACHWATCH badge is a rounded-rectangle nameplate with 6.4–6.5 mm letters, so it prints cleanly. The pockets are on the faces that touch the bed, so look from underneath in the slicer to see them



The three plates seen from under the bed: the white cut-outs are the badge pockets.



Where everything sits inside the shell.

Assembly

INSERTS: Heat-set inserts: soldering iron at about 220 °C for PETG (lower for PLA), insert tip if you have one. Set the insert on its hole and let it sink in under its own weight plus light pressure, keeping it straight. Stop when it's flush. Let it cool before screwing in.

C1	Flash SquachWatch (Step 0), then build and test the electronics (PART 1 + PART 2), MT3608 set to 5.0 V.	[ ]
C2	Press the 4 **M4 × 6 × 6 mm heat-set inserts** into the 4 corner bosses of the shell.	[ ]
C3	Glue the badges in with a few small drops of CA glue. Each fits only one way. **Horizontal:** both the Squachy badge and the SQUACHWATCH badge go on the back of the shell. **Vertical:** Squachy goes on the back, SQUACHWATCH on the bezel. (Plate 2: skip this.)	[ ]
C4	Put **Kapton tape** on the back of the CYD so its pins can never touch the battery or a board. Wrap the undersides of the TP4056 and MT3608 too.	[ ]

C5	Foam-tape the TP4056 into its pocket with its **USB-C pushed up to the charge-port wall, centred in the notch**. Test-fit your charging cable now: it should click fully in; add a dot of hot glue behind it so plugging in can't push it back.	[ ]
C6	Foam-tape the battery and the MT3608 into their rib pockets. LiPos swell slightly with age, so don't force anything tight.	[ ]
C7	Push the switch body against the far end wall with the lever through the slot; hot-glue it from inside.	[ ]
C8	Plug the power lead into **P1/P5** and fold its wires down into the gap at the USB end.	[ ]
C9	Lay the CYD on the 4 corner posts, screen up, USB end against the USB-end wall. It drops in between the 4 corner bosses. Check the USB ports line up with their window and the SD slot with its slot.	[ ]
C10	Fit the bezel and drive the 4 **M4 × 8 mm button-head screws** into the inserts with a 2.5 mm hex key. Snug, not tight.	[ ]
C11	Slide the stylus into its tube, tip first, from the switch end. Switch ON and confirm it boots. Done.	[ ]

Using it

Charging

1	**Charge through the TP4056's USB-C port**, the notch in the long wall under the stylus tube. The CYD's own USB port (on the short end) is only for flashing firmware; it does **not** charge the battery.	[ ]
2	**Best:** switch OFF, plug a USB-C charger into the TP4056. Wait for the "full" LED.	[ ]
3	You *can* run it while charging (switch ON), but the charger can't tell the battery's current from the CYD's current, so it may never show "full". That's fine for topping up; for a full charge, switch OFF.	[ ]
4	Charge time from empty: about 3.5 hours with a 1 A charger.	[ ]
5	Don't leave it at 0% for weeks; LiPos like being stored around half full.	[ ]

LED	Meaning
Charging LED on (usually red)	Charging
Full LED on (usually blue/green)	Full (or no battery detected)
Both off	No charger plugged in (normal)
Flickering	Bad battery connection or a weak charger; check B+/B– joints

Battery life

About 9–10 hours of continuous scanning with no add-ons connected. When the battery reaches ~2.5 V the TP4056's protection cuts power and the CYD switches off. Recharge it as soon as you can.

Flashing new firmware

Method	Do this
A (hard-wired)	Switch **OFF first**, then plug the computer into the CYD's USB port. Flash. Unplug. Switch on.
B (USB cable)	Switch OFF, unplug the internal cable from the CYD, plug the computer in. Flash. Swap back.

SquachWatch: on the web flasher, pick **2.8" CYD · ST7789**. If the screen stays white, untick 80 MHz and flash again; if still white, use **2.8" CYD · ILI9341**.

Troubleshooting

Symptom	Most likely cause	Fix
Nothing happens at switch ON	Switch in the OFF position; battery protection tripped; battery flat	Flip the switch. Plug in USB-C for 5 s to reset the protection. Charge.
0 V on TP4056 OUT+/OUT–	Protection latched after a short or deep discharge	Plug USB-C charger in briefly. If still 0 V, check B+/B– joints and Step 1.
Something gets HOT right away	Short circuit or reversed battery	**Switch OFF immediately.** Recheck B+/B– polarity and look for bridges.
MT3608 output far above 5 V	Trimmer not set	Disconnect the CYD at once. Redo Step 5.
MT3608 output doesn't change when turning	Trimmer at its end stop, or no input	Keep turning (up to 25 turns). Check VIN has battery voltage.
CYD boots then reboots, especially when scanning starts	Voltage too low under load; loose joint; battery nearly empty	Reset MT3608 to 5.05 V. Reflow VOUT joints. Charge.
Works on USB, not on battery (Method A)	Lead in the wrong connector, or VIN/GND swapped	Use P1 (P5 on dual-USB boards), not CN1/P3. Redo the VIN/GND meter test and Step A1 checks.
Works on USB, not on battery (Method B)	Cut-cable power wires reversed or a data wire used	Redo Step B2 with the meter.
Charge LED never goes to "full"	Charging with the switch ON	Switch OFF while charging.
Battery only lasts a few hours	Add-ons connected to CN1/P3; battery not fully charged	Unplug add-ons you don't need. Charge until the full LED.
Battery is puffy	Battery damaged	Stop using it. Recycle it. Replace.

Final checklist

1	Battery reads 3.6–4.2 V; red = + confirmed.	[ ]
2	Battery on B+/B–; OUT+/OUT– show battery voltage.	[ ]
3	Charges over USB-C; LED goes to "full".	[ ]
4	Switch cuts VIN+ on the MT3608; ON direction marked.	[ ]
5	MT3608 set to 5.00–5.10 V; screw locked with nail polish/glue.	[ ]
6	Method A: P1 lead verified (VIN/GND), TX/RX insulated. — or — Method B: cable power pair verified.	[ ]

7	Measured 5 V at the CYD end before plugging in; correct polarity.	[ ]
8	Boots SquachWatch on battery; nothing hot after 15 minutes.	[ ]
9	Unneeded add-ons unplugged from CN1/P3.	[ ]
10	Everything insulated with Kapton/heat-shrink; wires can't be pinched by the lid.	[ ]

Sources: CYD P1 pinout (VIN, TX, RX, GND) per espboards.dev and the ESPHome CYD notes · SquachWatch-CYD README and web flasher (squachwatch.com). Photos show the actual components used; wiring diagrams are drawn illustrations.

Credits: case design and this guide by **Dagromflomite** (MIT License; credit required). Squachy, the SquachWatch name and artwork: Talking Sasquach (talkingsasquach.com). Because the case includes these logos, it cannot be sold without express permission from Talking Sasquach. SquachWatch firmware: github.com/skizzophrenic/SquachWatch-CYD. Case files: https://github.com/DaGromflomite/squachwatch-cyd-case.